

Project Design Phase
Problem – Solution Fit

Date	03 October 2026
Team ID	SWTID-2026-5622
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	2 Marks

Problem – Solution Fit:

Educational institutions maintain student and faculty information such as names, roll numbers, branches, email addresses, and phone numbers. These records must be accessible only to authorized users.

Without proper access control, users may view or modify records that they are not permitted to access. General role-based permissions may not be sufficient when access must depend on a specific field value, such as the student's branch.

The system requires a secure mechanism to control record operations such as Read, Create, Write, and Delete based on user roles and field conditions.

Common problems include:

- Unauthorized viewing of institutional records.
- Lack of branch-specific record visibility.
- Uncontrolled record creation and modification.
- Unauthorized deletion of records.
- Insufficient separation of user and administrator permissions.

The proposed solution uses ServiceNow Access Control Lists (ACLs) and scripting to enforce role-based and field-based record access.

Customer / Target User :

Target User	Customer Problem / Need
Faculty Members	Need controlled access to institutional information.
ServiceNow Administrators	Need complete access to manage and maintain records.
Educational Institutions	Need confidentiality and controlled management of student records.

Problem–Solution Fit:

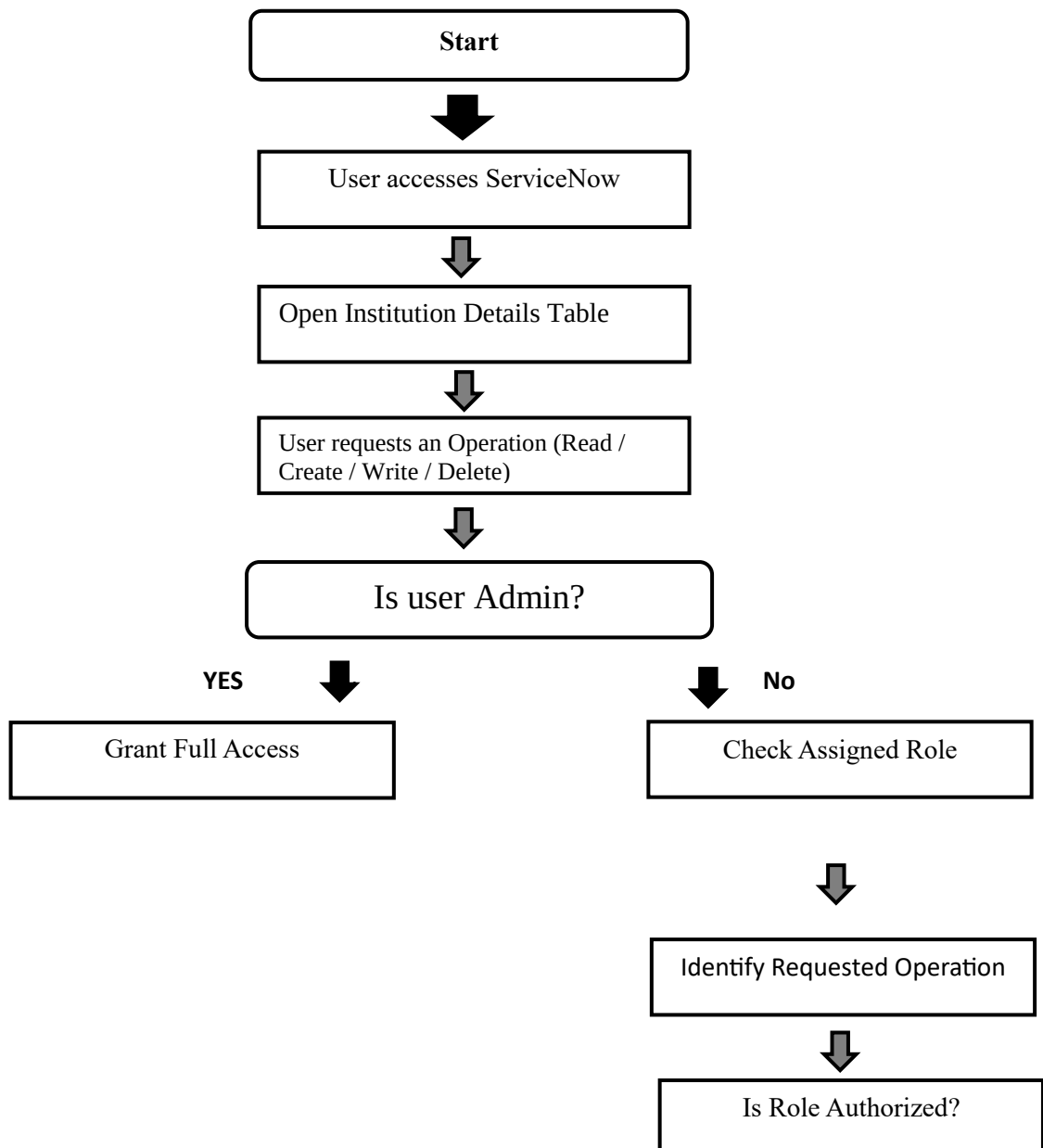
PS No.	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	EEE User	View EEE student records	I may access records from other branches	Record-level restrictions are required	Concerned about data security
PS-2	Faculty Member	Access authorized institutional data	Some records may be restricted	Access depends on roles and field values	Limited in accessing relevant information
PS-3	System Administrator	Manage all institutional records.	Regular users have restricted permissions.	Different users require different access levels.	Responsible for maintaining security.
PS-4	System User	Create or modify records	Unauthorized operations may occur	CRUD permissions are not properly separated	Concerned about record integrity
PS-5	Educational Institution	Protect student information	Unauthorized users may view or delete records	Field-based access control is required	Concerned about confidentiality

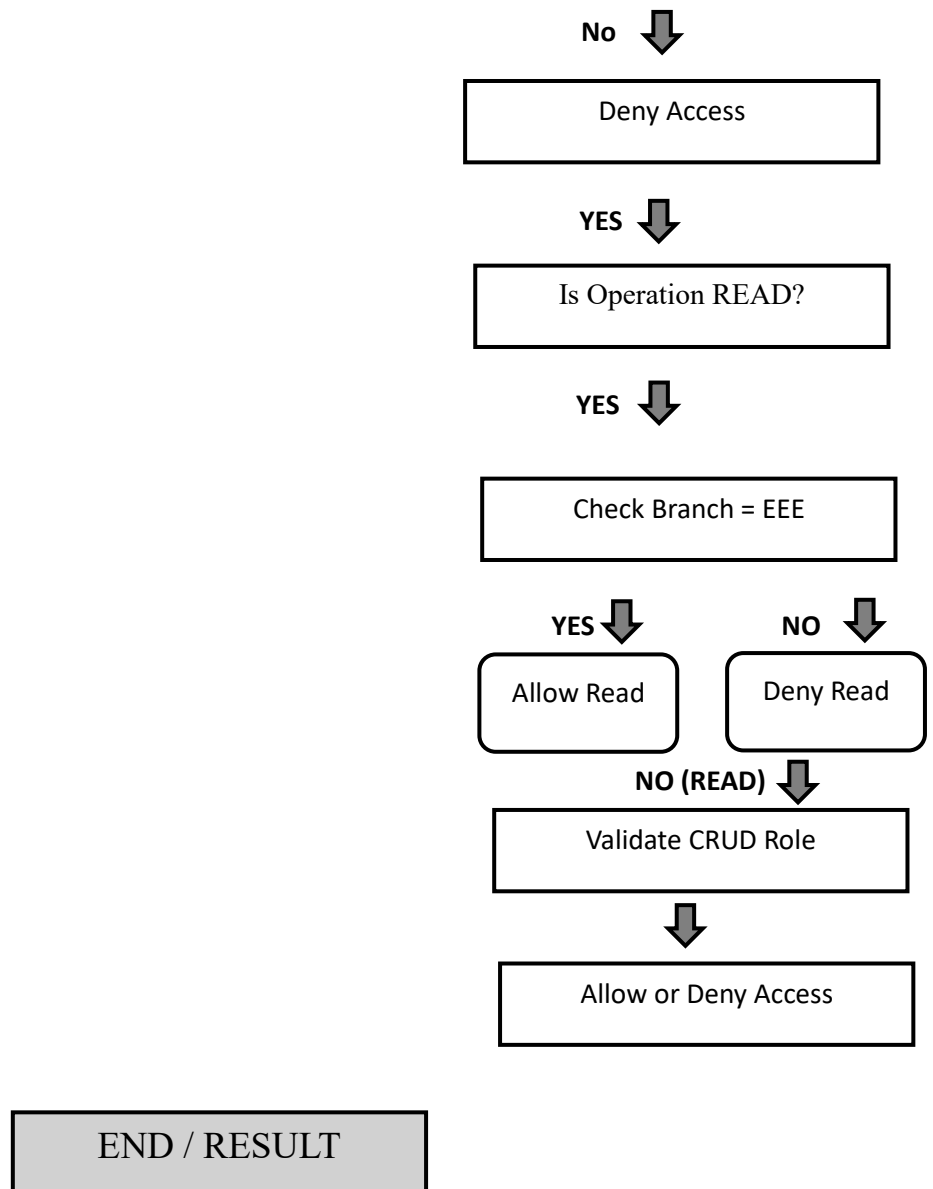
Existing Situation:

Aspect	Current Situation
Record Creation	Institutional records are stored in a custom ServiceNow table.
User Access	Users require assigned roles to perform operations.
Record Visibility	Branch-specific visibility requires additional access conditions.
Record Modification	Write permissions must be restricted to authorized users.
Administration	Administrators require full access for system management.

Proposed Solution:

Requirement	Proposed Solution
Secure record visibility	Configure ServiceNow Read ACL with role and branch conditions.
Branch-specific access	Allow authorized users to read records where Branch = EEE.
Controlled record creation	Configure Create ACL requiring role bb2.
Controlled record modification	Configure Write ACL requiring role bb3.
Controlled record deletion	Configure Delete ACL requiring role bb4.
Centralized security	Apply ACL rules to the custom table u_institution_details.

Proposed Solution Flow:



Complete End-to-End Process:

Step	Process
1	Create the custom table u_institution_details.
2	Configure student and faculty-related fields.
3	Create user roles bb1, bb2, bb3, and bb4.
4	Configure the Read ACL with the EEE branch condition.
5	Configure Create, Write, and Delete ACLs.
6	Add administrator authorization in the script.
7	Assign appropriate roles to test users.
8	Impersonate users to verify access permissions.
9	Confirm that unauthorized operations are denied.
10	Verify that administrators can access all records.

Problem–Solution Fit:

Problem	Solution	Fit
Unauthorized record viewing	Role and branch-based Read ACL.	Improved data confidentiality
Uncontrolled record creation	Create ACL with role bb2.	Controlled record creation
Unauthorized modification	Write ACL with role bb3	Improved data integrity
Unauthorized deletion.	Delete ACL with role bb4.	Protection against unauthorized deletion
Different access requirements.	Separate permissions for CRUD operations.	Granular access management